Module-1

Introduction: *Role of Mechanical Engineering in Industries and Society- Emerging Trends and Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.*

Introduction to Mechanical Engineering:

Mechanical Engineering is that branch of engineering which combines Science, mathematics, engineering principles with materials science in analyzing, designing, manufacturing maintenance and testing of mechanical systems.

There is a wide scope after completion of this course. Mechanical Engineers are instrumental in designing efficient motor vehicles, aircrafts, manufacturing units and industrial machinery and contribute towards the development of complex machinery systems, engines and power equipment.

Role Of Mechanical Engineering in Industries and Society

Role of Mechanical Engineering in Industries Knowledge of Mechanical Engineering in the industry helps to do the following:

- Analyze problems to see how mechanical and thermal devices might help solve a particular problem
- Design or redesign mechanical and thermal devices or subsystems, using analysis and computer-aided design
- Develop and test prototypes of devices they design
- Investigate equipment failures or difficulties to diagnose faulty operation and to recommend remedies
- Analyze the test results and change the design or system as needed
- Oversee the manufacturing process for the device

The following are the examples of Role of Mechanical Engineering in Industries:

- **Manufacturing Industry**: Mechanical engineers are involved in the design and optimization of manufacturing processes and equipment. They work on improving productivity, quality, and cost-effectiveness in production lines. They also contribute to automation and robotics, implementing advanced manufacturing technologies.

- **HVAC and Building Systems**: Mechanical engineers play a significant role in designing **H**eating **V**entilation and **A**ir **C**onditioning (HVAC) systems for buildings. They ensure energy efficiency, comfort, and air quality. They

also work on plumbing systems, fire protection, and other mechanical systems in buildings.

- **Robotics and Automation**: Mechanical engineers contribute to the development of robotic systems and automation technologies. They design robotic mechanisms, grippers, and control systems. They work on applications in industries such as manufacturing, healthcare, logistics, and agriculture.

- **Consumer Products**: Mechanical engineers are involved in the design and development of various consumer products, including appliances, electronic devices, sporting equipment, and more. They ensure product functionality, safety, and user experience.

- **Engineering**: Mechanical engineers contribute to the design and development of medical devices and equipment. They work on prosthetics, implants, diagnostic tools, and rehabilitation devices. They collaborate with healthcare professionals to improve patient care and quality of life.

Role of Mechanical Engineers in Society

The important role of Mechanical Engineers in Society is listed below:

- **Transportation**: Mechanical engineering is largely concerned with road transportation, and networks, rail and subway systems, airports, and shipping ports.

- **Medical**: A mechanical engineer specialized in medical science can be called as 'medical engineer'. Engineer's responsibilities in medicine can include research, development, testing and evaluation of medical devices, advisement on new biomedical purchases for hospitals and medical centers.

- **Agriculture**: The graduates are trained professionals who understand technology and can solve problems in the areas of agricultural production and management of rural areas with a special focus on machinery.

- **Defense:** Manage the teams of skilled technicians that monitor, maintain and repair formidable military hardware such as tanks, artillery guns and armored logistic vehicles. Not all types of entry are open for this job.

- **Power generation**: When working in thermal power plants, mechanical engineers make sure heavy machinery like boilers and turbines, are working in optimal condition and power is continually generated. Mechanical engineers also work with the operations of the plant.

Emerging Trends and Technologies in Different Sector

Some of important emerging trends and technologies in different sector are discussed below:

- **ENERGY SECTOR**
    1. **Renewable Energy**: There is a growing focus on renewable energy sources such as solar, wind, hydro, and geothermal power. Advancements in solar panel efficiency, wind turbine technology, and energy storage systems are driving the adoption of renewable energy on a larger scale.

    2. **Energy Storage:** The development of advanced energy storage technologies, such as lithium-ion batteries, flow batteries, and hydrogen storage, is crucial for managing the intermittent nature of renewable energy sources and enabling grid flexibility.

    3. **Smart Grids**: Smart grid technologies incorporate digital communication and control systems into traditional electrical grids. They enable more efficient electricity distribution, demand response mechanisms, and integration of renewable energy sources into the grid.

- **MANUFACTURING SECTOR**

    1. **Additive Manufacturing (3D Printing):** Additive manufacturing allows the production of complex parts and prototypes using layer-by-layer material deposition. It offers advantages such as design flexibility, reduced material waste, and rapid prototyping, transforming manufacturing processes.
    2. **Industrial Internet of Things (IIoT):** IIoT involves the integration of sensors, devices, and data analytics to optimize manufacturing processes, monitor equipment performance, enable predictive maintenance, and enhance overall productivity.
    3. **Artificial Intelligence (AI) and Machine Learning**: AI and machine learning are being used to improve automation, optimize production lines, and analyze vast amounts of data for predictive analytics and quality control in manufacturing.

- **AUTOMOTIVE SECTOR**
    1. **Electric Vehicles (EVs):** The automotive industry is experiencing a shift towards electric vehicles powered by batteries or hydrogen fuel cells, reducing dependence on fossil fuels and minimizing emissions.
    2. **Autonomous Vehicles**: Self-driving or autonomous vehicles are being developed with advanced sensors, machine learning algorithms, and

connectivity technologies. They have the potential to revolutionize transportation, improving safety, efficiency, and mobility.

3. **Connected Cars**: Connectivity and communication technologies are being integrated into vehicles, enabling features such as real-time traffic information, remote diagnostics, and vehicle-to-vehicle communication.

- **AEROSPACE SECTOR**
    1. **Lightweight Materials**: The aerospace industry is exploring advanced lightweight materials, such as carbon fiber composites and titanium alloys, to reduce weight and improve fuel efficiency in aircraft.
    2. **Electric Propulsion**: Electric propulsion systems, including electric motors and batteries, are being developed for aircraft. They offer potential benefits such as reduced emissions, noise, and operating costs.
    3. **Advanced Aerodynamics**: Innovations in aerodynamics, including improved wing designs, laminar flow control, and adaptive structures, aim to enhance aircraft performance and fuel efficiency

- **MARINE SECTOR**

    1. **Alternative Fuels**: The marine industry is exploring alternative fuels like liquefied natural gas (LNG), biofuels, and hydrogen to reduce emissions and comply with environmental regulations.
    2. **Autonomous Vessels**: Autonomous or unmanned vessels are being developed for various applications, including cargo shipping, surveying, and offshore operations. They have the potential to increase efficiency, reduce costs, and enhance safety.
    3. **Sustainable Ship Design**: The focus on sustainability in the marine sector includes the development of eco-friendly ship designs, such as energy-efficient hull shapes, hybrid propulsion systems, and improved waste management systems.

ENERGY

Energy is defined as the capacity to do work, Energy exists in various forms such as mechanical energy, electrical energy, thermal energy and chemical energy.

Various sources of energy:

(i) Fossil fuels, (ii) Atomic fuels (nuclear energy), (iii) Flow of river water (hydro energy), (iv) Sun (solar energy), (v) Wind, (vi) Tides, (vii) Earth (geothermal energy), (viii) Bio-mass, etc.

Classifications of Energy Sources:

The sources of energy are classified in the following two ways:

(a) Conventional and non- conventional sources of energy
(b) Renewable and non-renewable sources of energy

A) Conventional and non- conventional sources of energy

Conventional energy sources OR commercialized energy sources: Conventional sources of energy are the natural energy resources which are regularly used for many years and are accepted as fuel to produce heat, light, food and electricity.

eg.: Fossil fuel, Hydel energy, Nuclear energy

Non-conventional energy sources OR Non commercialized energy sources: These energy sources achieved popularity in recent years and their extraction process not fully developed.
These energy sources and will provide energy is in dilute form.
eg. solar energy, wind energy, tidal energy, ocean thermal energy, geothermal energy, bio energy, fuel cells, solid wastes, hydrogen energy etc.

B. Renewable and non-renewable sources of energy

Renewable sources of energy: The renewable sources of energy are defined as the energy sources which are continuously produced in nature and are essentially inexhaustible or non-depletable. These energy sources replenish themselves naturally in a relatively short time and therefore will always be available.

eg. : Solar energy, wind energy, tidal energy, hydel energy, ocean thermal energy, bio energy, geothermal energy, peat, fuel wood, fuel cells, solid wastes, hydrogen energy etc.

Non-renewable sources of energy: The non-renewable energy sources are those natural resources that are available in limited quantity and these resources cannot be renewed in short durations.
eg.: Fossil fuels (coal, petroleum and petroleum products), nuclear fuels and heat traps
Biofuels

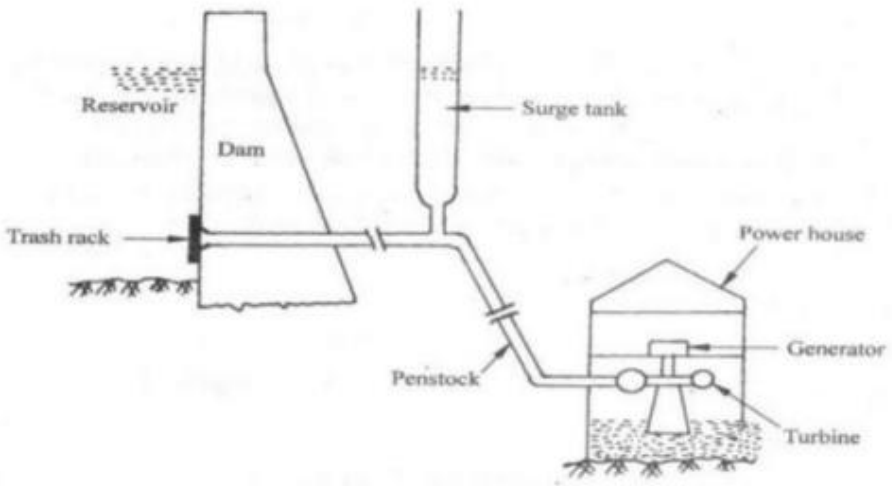
Difference between Conventional and non- conventional sources of energy:

Conventional sources of energy	Non- conventional sources of energy
These are widely used and economical	These are rarely used and initial cost is high
Most of them are exhaustible	Most of them are in-exhaustible
Most of them are pollute the environment	Most of them are environment friendly
They are reliable (continuous supply of energy is possible)	They are not reliable (continuous supply of energy is not possible)
Energy transmission cost is high	Energy transmission cost is low
Example: Fossil fuels, Hydel energy	Example: solar, wind, tidal etc.

Difference between Renewable and non-renewable sources of energy:

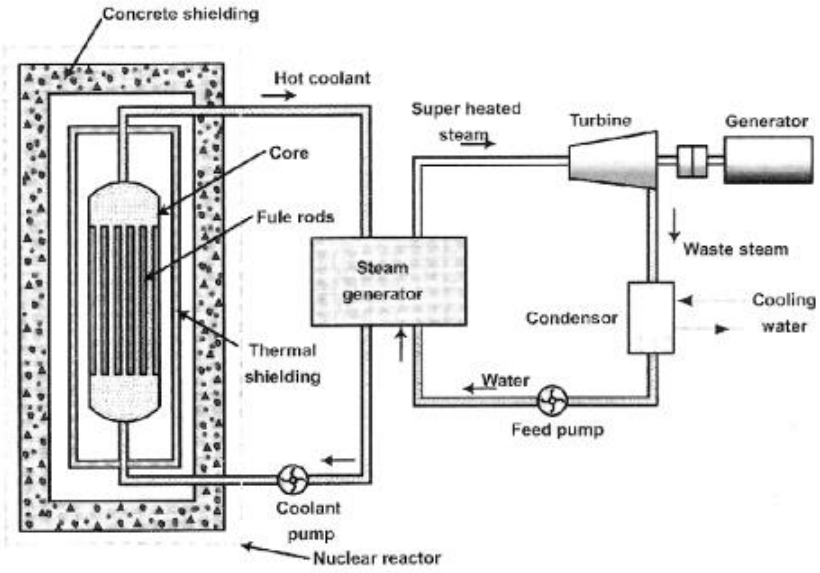
Renewable sources of energy	Non- Renewable sources of energy
These are in-exhaustible	These are exhaustible
Freely available and environment friendly	Not freely available & hazardous to environment
Initial cost is high but maintenance cost is low	Initial cost is low but maintenance cost is high
Energy concentration varies from region to region	Energy concentration almost same in all region
Example: solar, hydel, wind, tidal etc.	Example: Fossil fuels, nuclear fuels.

Hydro-Energy:



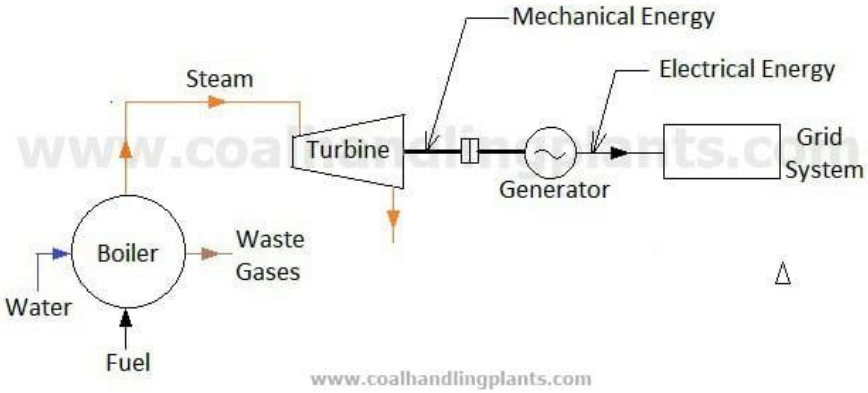
Hydroelectric power (often called hydropower) is considered as a conventional and a renewable energy source. By constructing a dam across the river water will be stored in the reservoir at different heights. Hence potential energy will be increased. Now the water is released through a penstock, to drive the turbine. At the tip of the penstock pipe water velocity will be increased due to the nozzle hence kinetic energy is increased. The flowing water causes the turbine to rotate, converting the water 's kinetic energy into mechanical energy. The mechanical energy produced by the turbine is converted into electric energy using a turbine generator.

Nuclear energy:



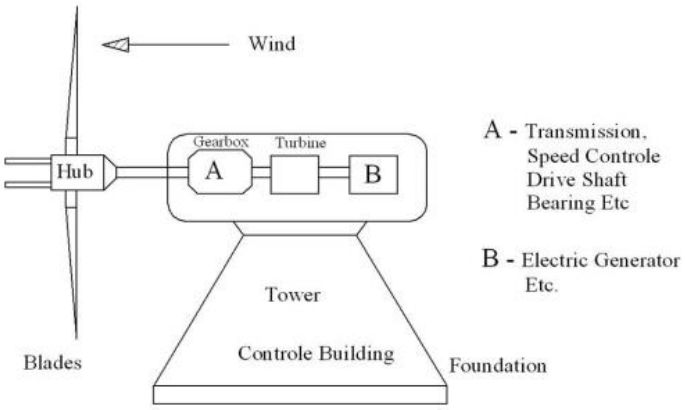
Nuclear energy is the energy released during nuclear fission or fusion, especially when used to generate electricity. The most common type of nuclear fuel is fissile elements that can be made to undergo nuclear fission chain reactions in a nuclear reactor. The most common nuclear fuels are 235U and 239Pu. The basic nuclear generating station energy cycle is shown in the above figure. Fuel containing fissile material (Uranium) is fed to the reactor where fission takes place. The energy liberated appears in the form of heat, which is used to boil water in Heat exchanger or steam generator. The steam produced from the boiling water spins a turbine-generator set, where the heat is converted first to kinetic energy and then it will convert to mechanical energy in the turbine and to electricity by the generator.

Fossil fuels: which include coal, oil, and natural gas, are a primary source of energy for electricity generation, transportation, and various industrial and residential applications. These non-renewable resources are formed from the remains of ancient plants and animals over millions of years. When burned, fossil fuels release stored chemical energy in the form of heat, which can be used to produce mechanical work, heat, or electricity.



In thermal power plants, the fossil fuel is used to heat the water and convert it into high pressure steam. When this high-pressure steam flows over the turbine blades it makes the turbine shaft to rotate. Steam turbines are used in thermal plants. The generator will convert mechanical energy from a shaft to electrical energy.

Wind energy:



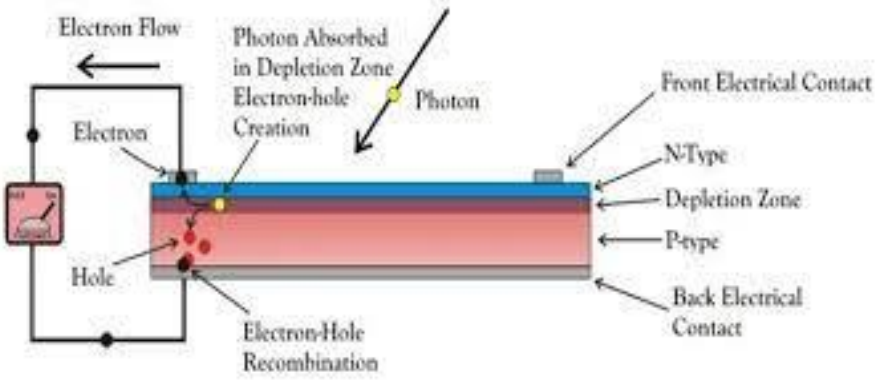
which produces mechanical energy by rotating the wind turbine shaft this intern coupled to the electric generator which generates electricity.

Wind energy conversion devices are commonly known as wind turbines. The wind turbine consists of a number of vanes or blades radiating from a hub or central axis. When the wind blows against the vanes of a wind turbine, rotational motion is caused in the wind turbine. Thus, the kinetic energy of wind is converted into mechanical work. This mechanical work can be used directly to run a machine or to run a generator to produce electricity. Wind energy can be utilized in places wind velocity is considerably high.

Solar Energy:

Solar energy refers to the energy produced and radiated by the sun. Solar energy can be converted directly or indirectly into other forms of energy, such as heat and electricity which can be utilized by man
This heat may then be used in various ways, which may be divided broadly into two classes as (a) Direct thermal applications: Helio-electrical and Helio-thermal process (b) Indirect thermal applications: Biomass energy (Helio-chemical), wind energy, tidal energy, ocean thermal energy etc.

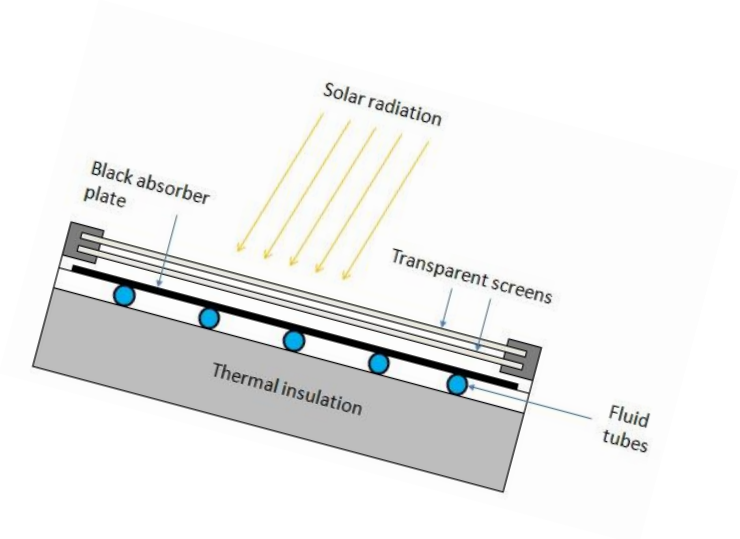
Helio-electrical Process:



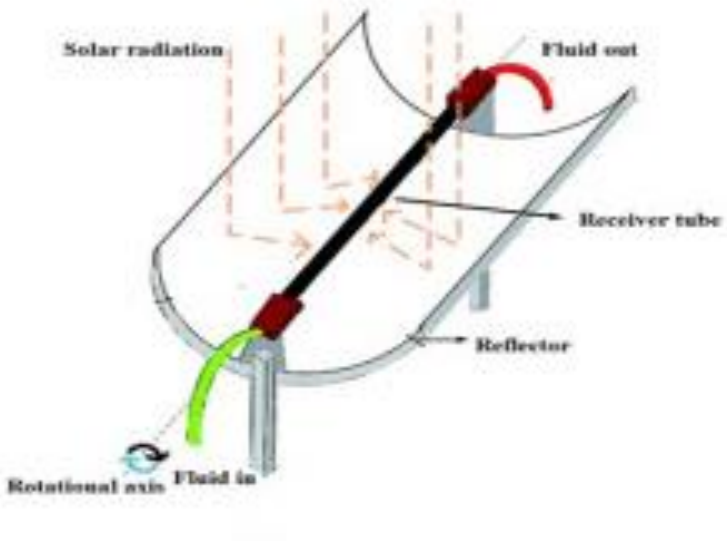
A photovoltaic (PV) cell, also known as a solar cell, is a semiconductor device that converts light energy into electrical energy. It works by utilizing the photovoltaic effect, where photons from sunlight are absorbed by the semiconductor material, causing electrons to be released and creating an electric current.
Sunlight, composed of photons, strikes the PV cell. When a photon is absorbed by the semiconductor material (usually silicon), it transfers its energy to an electron, causing the electron to jump to a higher energy level and become free. The freed electron (negatively charged) and the remaining "hole" (positively charged) are separated due to the electric field within the cell, typically created by a p-n junction. The separated electrons flow through an external circuit, creating an electrical current, while the holes flow in the opposite direction. This flow of electrons results in a direct current (DC). The generated DC electricity can then be used directly or converted to alternating current (AC) for wider use.

Helio-Thermal Process:
In flat plate collectors the incident solar radiation is absorbed by the collectors surface itself, which are usually coated with black paint (usually electroplated), covered with transparent glass cover on top and insulated all around to prevent the heat loss from the collector surface. The black collector surface gets heated up and then in turn transfers the heat to the

fluid passing through the tubes which are either welded or soldered or are integral part of the collector plate.



Flat plate collectors are usually sloped and oriented in one particular direction and are capable of collecting both diffuse and beam radiation. Since there are no moving parts in it, the repair and maintenance cost is also nil or negligible. A maximum of 100°C can be easily achieved using flat plate collectors and are more popularly used in solar water heating applications and solar air heaters as they are relatively cheaper as compared to the cost of concentrating collectors.



In concentrating collectors, the incident solar radiation falls on a large curved surface from where it is reflected and focused on to focal point or line depending upon the type of the geometrical construction of the concentrating collector.

Helio-chemical Process: The most important chemical reaction on the earth is the reaction of sunlight and green plants. Radiant energy of the sun is absorbed by the green pigment chlorophyll in the plant and is stored within the plant in the form of chemical bond energy.

1. The process of photosynthesis has two main steps: (i) Splitting of H_2O molecules into H_2 and O_2 under the influence of chlorophyll and sunlight. This phase of reaction is called the light-reaction. In this reaction, light absorbed by chlorophyll causes photolysis of water. O_2 escapes and H_2 is transformed into some unknown compounds. The solar energy is converted into potential chemical energy.
2. In the second step, hydrogen is transferred from this unknown compound to CO_2 to form starch or sugar. Formation of starch or sugar is dark reaction not requiring sunlight.

Biofuels are energy sources made from living things, or the waste that living things produce. It is the renewable energy source. Biofuels are combustible fuels created from biomass. There are two main types of biofuels one is **ethanol** and another **biodiesel**.

The heat generated from combustion is used to To produce steam, which turns turbines in power plants, generating electricity.

Ethanol

Ethanol can be produced by the action of microorganisms and enzymes through the fermentation of sugar cane. Ethanol can be used in petrol engines as a replacement for gasoline; it can be mixed with gasoline to any percentage. Most existing car petrol engines can run on blends of up to 15% bioethanol with petroleum/gasoline.

Advantages:

• Unlike petroleum, ethanol is a renewable resource

• Ethanol burns more cleanly in air than petroleum, producing less carbon (soot) and carbon monoxide

• The use of ethanol as opposed to petroleum could reduce carbon dioxide emissions, provided that a renewable energy resource was used to produce crops required to obtain ethanol and to distil fermented ethanol

Disadvantages:

• Ethanol has a lower heat of combustion (per mole, per unit of volume, and per unit of mass) that petroleum

• Large amounts of arable land are required to produce the crops required to obtain ethanol, leading to problems such as soil erosion, deforestation, fertilizer run-off and salinity.

• Major environmental problems would arise out of the disposal of waste fermentation liquors.

• Typical current engines would require modification to use high concentrations of ethanol

Biodiesel

It is produced from oils or fats using transesterification and is a liquid similar in composition to fossil/mineral diesel. Chemically, it consists mostly of fatty acid methyl (or ethyl) esters (FAMEs). Feedstocks for biodiesel include animal fats, vegetable oils, soyabean, jatropha, mahua, mustard, flax, sunflower, palm oil, hemp, field pennycress, Pongamia pinnata and algae. Pure biodiesel (B100) currently reduces emissions with up to 60% compared to diesel.

Advantages of biodiesel fuel:

• An excessive production of soybeans in the world makes it an economic way to utilize this surplus for manufacturing the Biodiesel fuel.
• One of the main biodiesel fuel advantages is that it is less polluting than petroleum diesel.
• The lack of sulfur in 100% biodiesel extends the life of catalytic converters.
• Another of the advantages of biodiesel fuel is that it can also be blended with other energy resources and oil.
It can also be distributed through existing diesel fuel pumps, which is another biodiesel

fuel advantage over other alternative fuels.
• The lubricating property of the biodiesel may lengthen the lifetime of engines.

Disadvantages of biodiesel fuel

i) At present, Biodiesel fuel is bout one and a half times more expensive than petroleum Diesel fuel.
ii) It requires energy to produce biodiesel fuel from crops
iii) As Biodiesel cleans the dirt from the engine, this dirt can then get collected in the fuel filter, thus clogging it. So, filters must be changed after the first several hours of biodiesel use.
iv) Biodiesel fuel distribution infrastructure needs improvement.

Application of Biofuels

i) As an alternative fuel for automobiles
ii) ii) Cooking purpose
iii) Lubrication in place of conventional oils
iv) In Power plants to generate electricity
v) In marine industry to run the ships

Environmental Issues

Environmental issues are harmful effects of human activity on the biophysical environment. The various environmental issues are global warming, ozone layer depletion, air pollution, water pollution, natural resource depletion, waste disposal. Climatic changes, deforestation, acid rain etc.

Green House Effect: The greenhouse effect is a natural process that warms the Earth's surface. When the Sun's energy reaches the Earth's atmosphere, some of it is reflected to

space and the rest is absorbed and re-radiated by greenhouse gases. Greenhouse gases include water vapor, carbon dioxide, methane, nitrous oxide, ozone and some artificial chemicals such as chlorofluorocarbons (CFCs). The absorbed energy warms the atmosphere and the surface of the Earth.

Step 1: Solar radiation reaches the Earth's atmosphere - some of this is reflected into space.
Step 2: The rest of the sun's energy is absorbed by the land and the oceans, heating the Earth.
Step 3: Heat radiates from Earth towards space.
Step 4: Some of this heat is trapped by greenhouse gases in the atmosphere, keeping the Earth warm enough to sustain life.
Step 5: Human activities such as burning fossil fuels, agriculture and land clearing are increasing the amount of greenhouse gases released into the atmosphere.
Step 6: This is trapping extra heat and causing the Earth's temperature to rise.

Global warming

The gradual heating of Earth's surface, oceans and atmosphere, is caused by human activity, primarily the burning of fossil fuels that pump carbon dioxide (CO_2), methane and other greenhouse gases into the atmosphere. Global warming is the long-term rise in the average temperature of the Earth's climate system

Causes for Global Warming:

Global warming occurs when carbon dioxide (CO_2) and other air pollutants and greenhouse gases collect in the atmosphere and absorb sunlight and solar radiation that have bounced off the earth’s surface. Normally, this radiation would escape into space—but these pollutants, which can last for years to centuries in the atmosphere, trap the heat and cause the planet to get hotter. That's what's known as the greenhouse effect.

Effects of Global warming

Ocean Warming, Ice Loss, change in climate, Sea level rise, Depletion of ozone layer, Causes for unseasonal rains, Causes for different types of skin diseases

Ozone Layer Depletion:

The ozone layer is an invisible layer of protection around the planet that protects us from the Sun’s harmful rays (Ultra-Violet Rays). In other words, it acts as nature’s umbrella against UV radiations.
Ozone depletion is the thinning of earth’s ozone layer. This is caused due to the presence of greenhouse gases in the atmosphere. Depletion of the ozone layer is mainly due to the interaction of ozone with chlorine and bromide present in Chlorofluorocarbons (CFC’s). Once these toxic gases reach the upper atmosphere, they cause a hole in the ozone layer causing its depletion. This is one of the most important current environmental challenges.

Effects of Ozone Layer Depletion:

- Increases the flux of UV radiations over earth’s biosphere
- Causes skin cancer and eyes disorders
- Decreases the rate of photosynthesis
- Leads to greenhouse effect
- It causes decrease in the yield of crops
- Adversely affects human immunity system and causes genetic abnormalities
- Causes the degradation of plaints, plastics and other polymer materials
- It affects the marine life